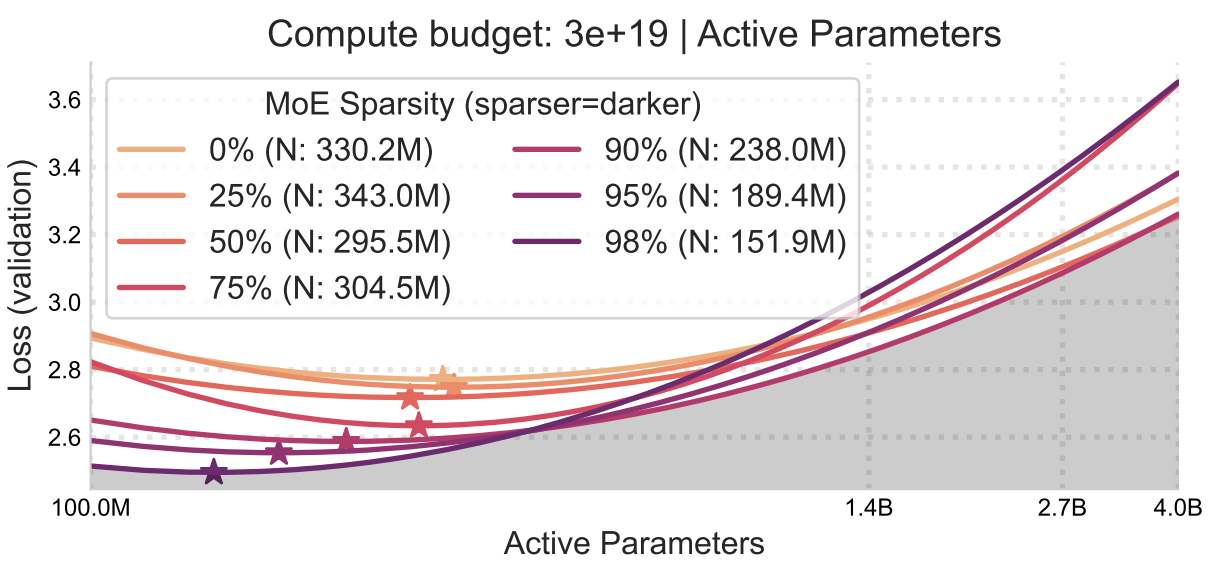
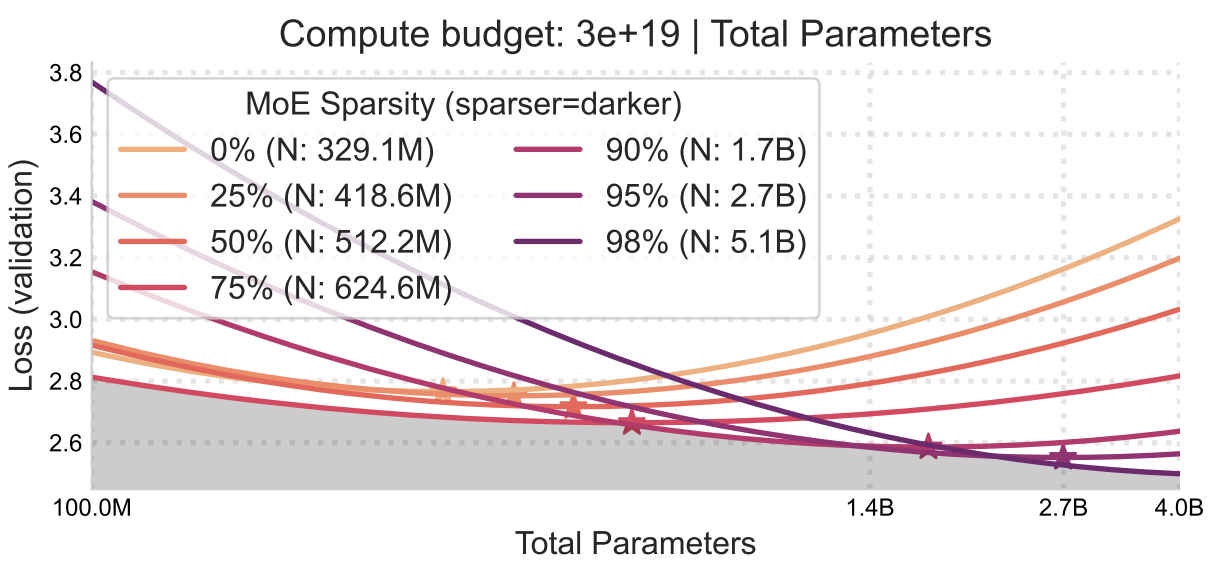
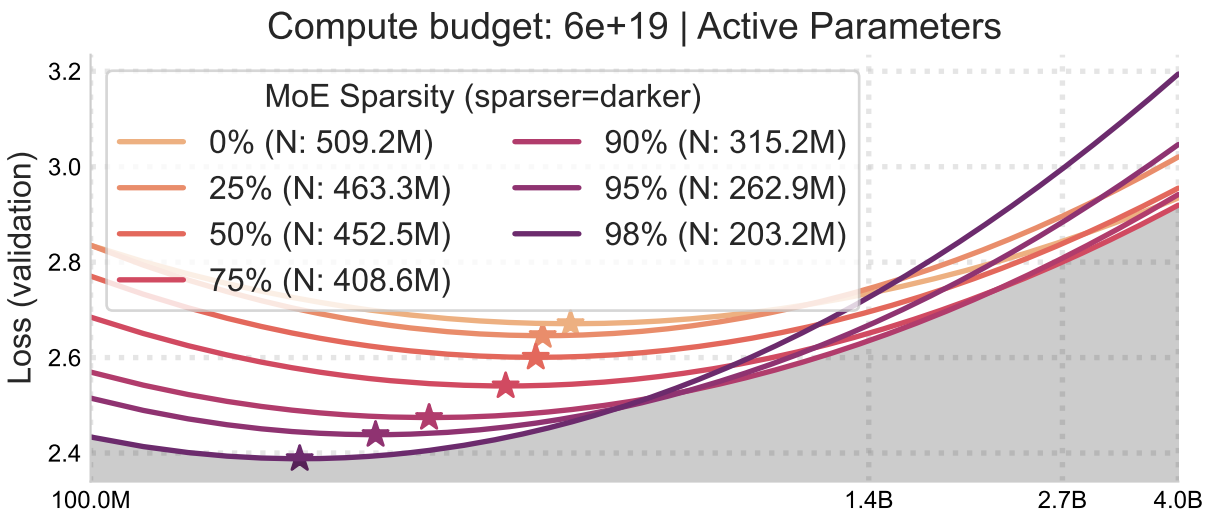
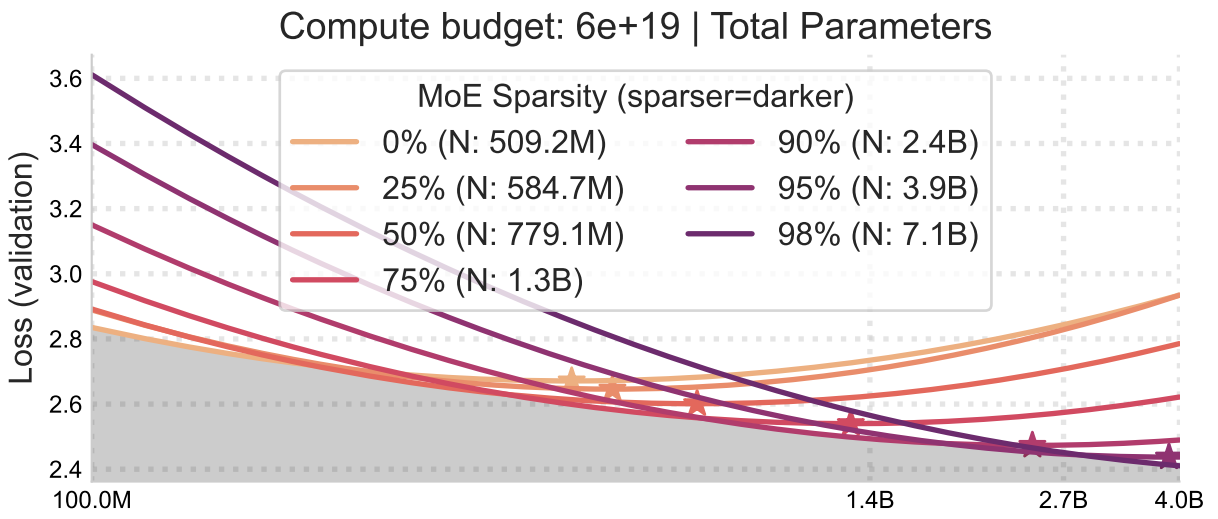
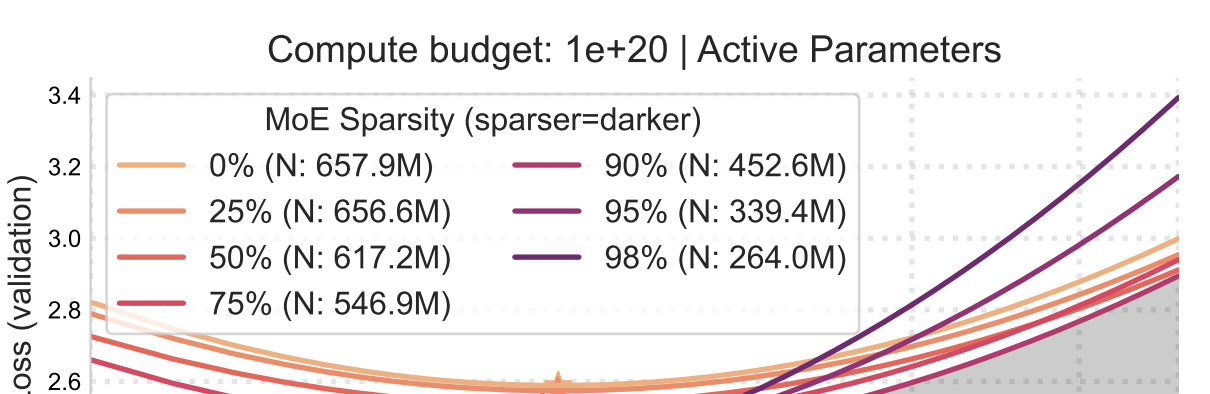
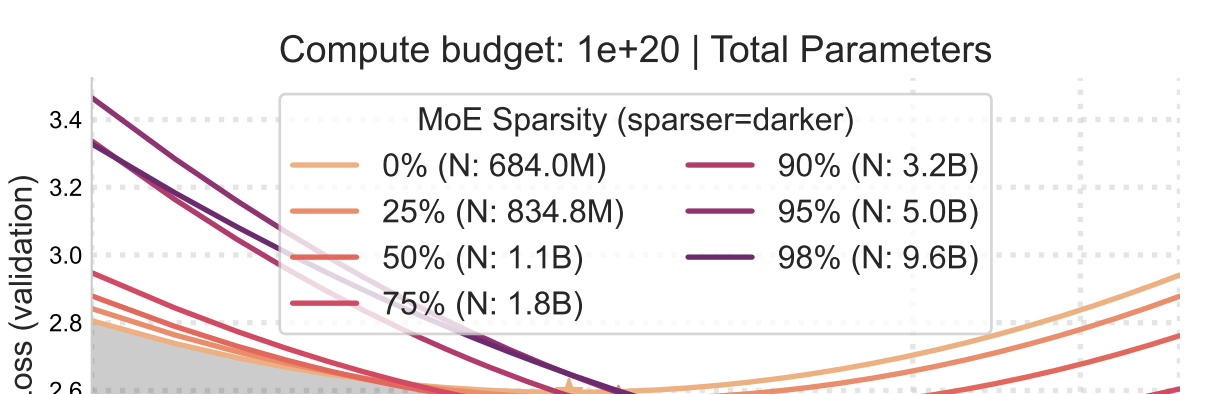

Compute budget: 3e+19 | Active Parameters
MoE Sparsity (sparser=darker)
0% (N: 330.2M)
25% (N: 343.0M)
50% (N: 295.5M)
75% (N: 304.5M)
90% (N: 238.0M)
95% (N: 189.4M)
98% (N: 151.9M)
Loss (validation)
3.6
3.4
3.2
3.0
2.8
2.6
100.0M
1.4B
2.7B
4.0B
Active Parameters

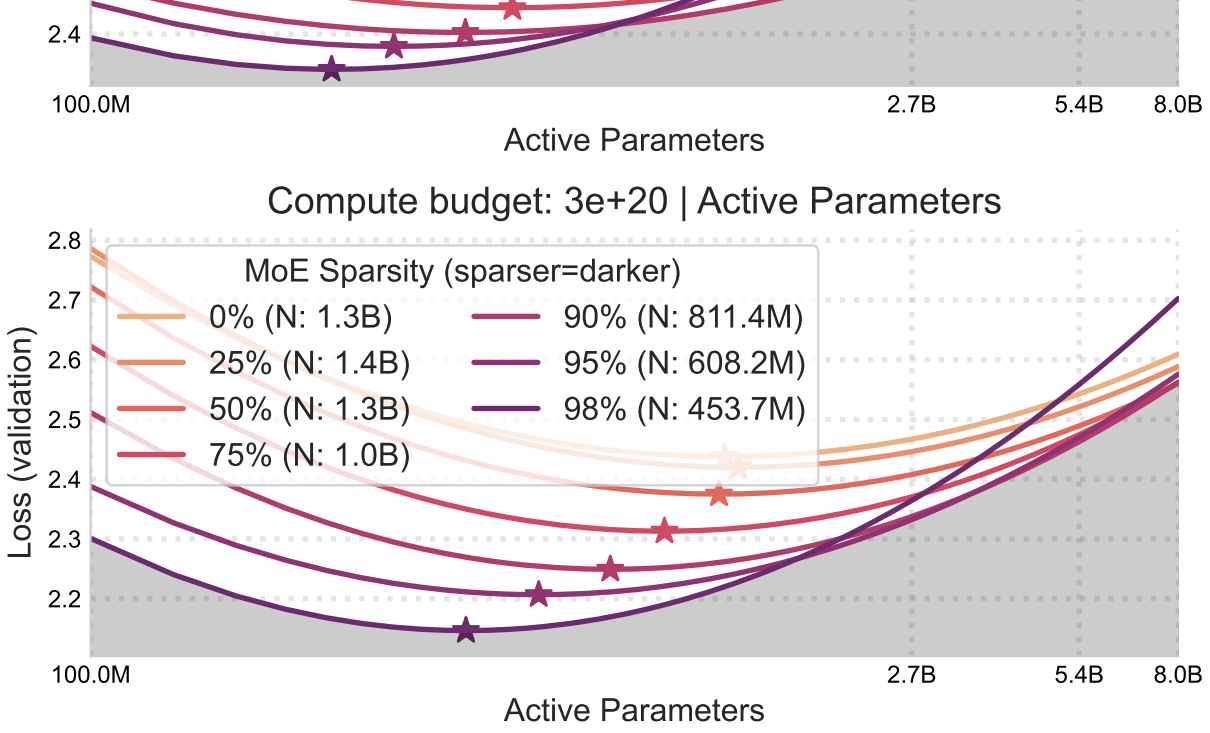

Compute budget: 3e+19 | Total Parameters
MoE Sparsity (sparser=darker)
0% (N: 329.1M)
25% (N: 418.6M)
50% (N: 512.2M)
75% (N: 624.6M)
90% (N: 1.7B)
95% (N: 2.7B)
98% (N: 5.1B)
Loss (validation)
3.8
3.6
3.4
3.2
3.0
2.8
2.6
100.0M
1.4B
2.7B
4.0B
Total Parameters

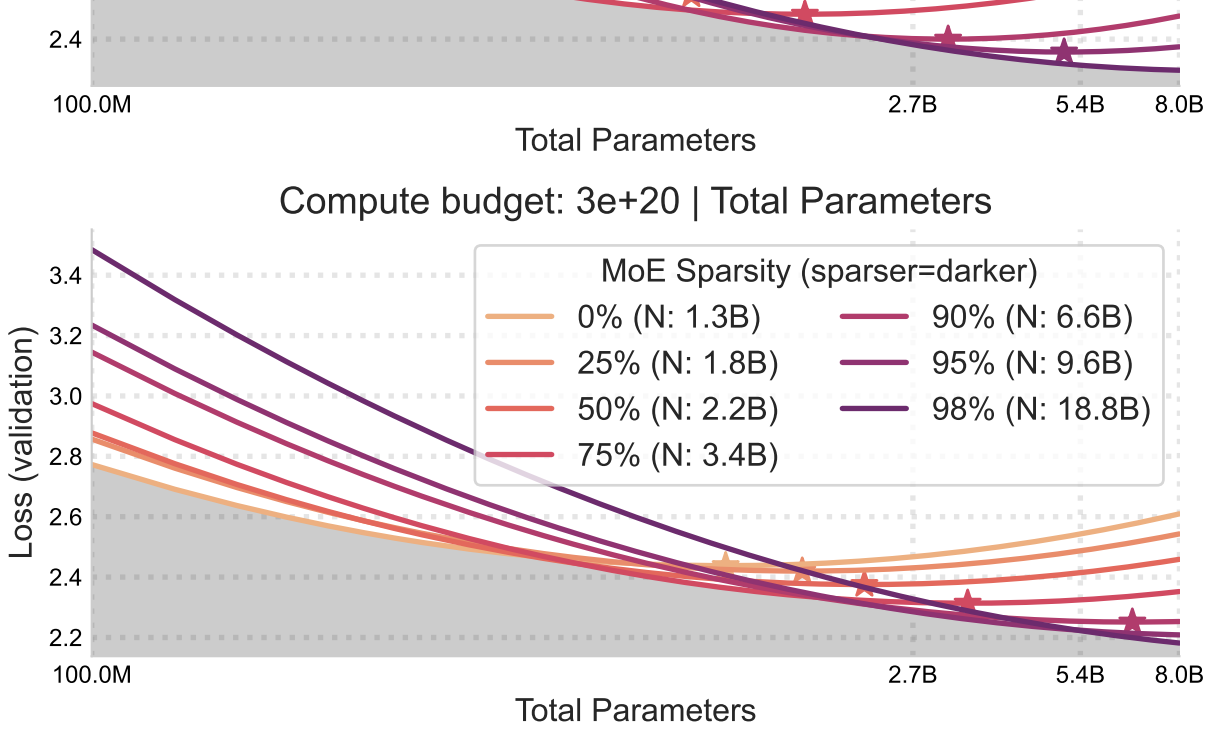

Compute budget: 6e+19 | Active Parameters
MoE Sparsity (sparser=darker)
0% (N: 509.2M)
25% (N: 463.3M)
50% (N: 452.5M)
75% (N: 408.6M)
90% (N: 315.2M)
95% (N: 262.9M)
98% (N: 203.2M)
Loss (validation)
3.2
3.0
2.8
2.6
2.4
100.0M
1.4B
2.7B
4.0B
Active Parameters

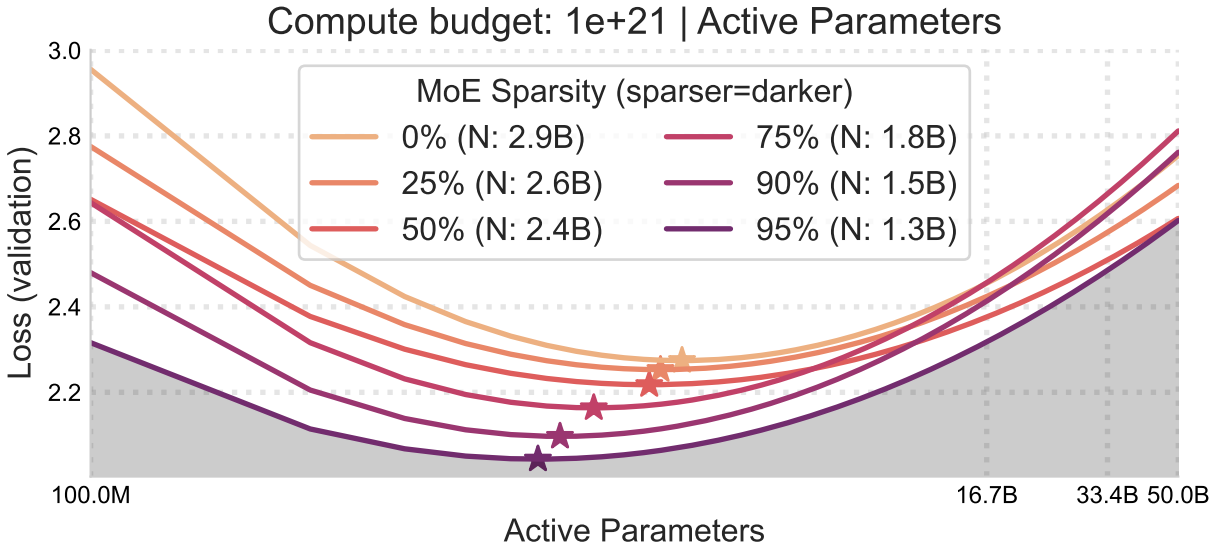

Compute budget: 6e+19 | Total Parameters
MoE Sparsity (sparser=darker)
0% (N: 509.2M)
25% (N: 584.7M)
50% (N: 779.1M)
75% (N: 1.3B)
90% (N: 2.4B)
95% (N: 3.9B)
98% (N: 7.1B)
Loss (validation)
3.6
3.4
3.2
3.0
2.8
2.6
2.4
100.0M
1.4B
2.7B
4.0B
Total Parameters

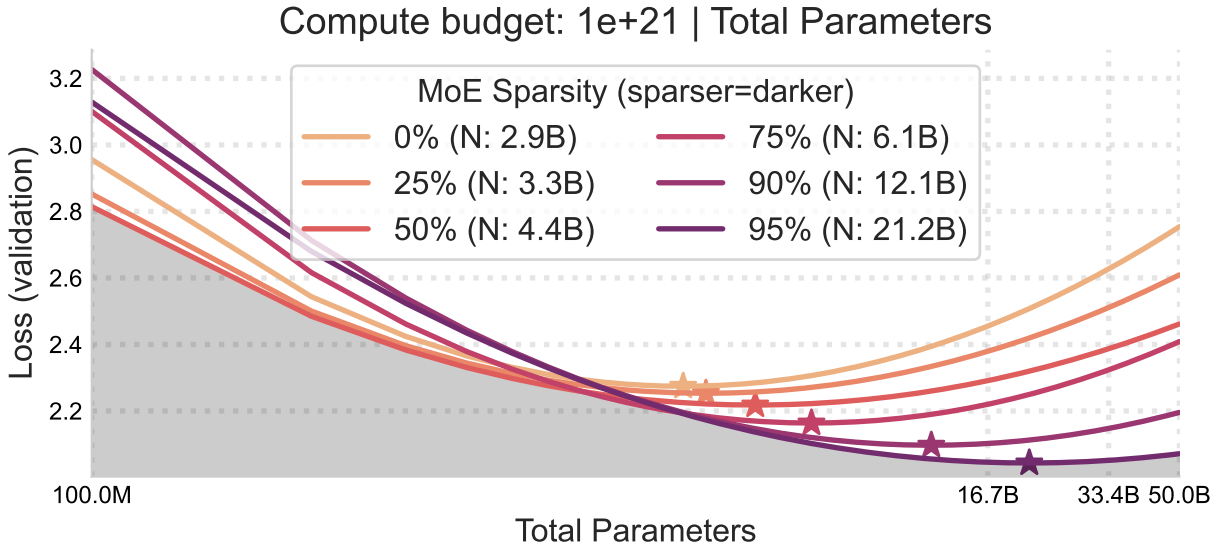

Compute budget: 1e+20 | Active Parameters
MoE Sparsity (sparser=darker)
0% (N: 657.9M)
25% (N: 656.6M)
50% (N: 617.2M)
75% (N: 546.9M)
90% (N: 452.6M)
95% (N: 339.4M)
98% (N: 264.0M)
Loss (validation)
3.4
3.2
3.0
2.8
2.6
2.4
100.0M
2.7B
5.4B
8.0B
Active Parameters


Compute budget: 1e+20 | Total Parameters
MoE Sparsity (sparser=darker)
0% (N: 684.0M)
25% (N: 834.8M)
50% (N: 1.1B)
75% (N: 1.8B)
90% (N: 3.2B)
95% (N: 5.0B)
98% (N: 9.6B)
Loss (validation)
3.4
3.2
3.0
2.8
2.6
2.4
100.0M
2.7B
5.4B
8.0B
Total Parameters


Compute budget: 3e+20 | Active Parameters
MoE Sparsity (sparser=darker)
0% (N: 1.3B)
25% (N: 1.4B)
50% (N: 1.3B)
75% (N: 1.0B)
90% (N: 811.4M)
95% (N: 608.2M)
98% (N: 453.7M)
Loss (validation)
2.8
2.7
2.6
2.5
2.4
2.3
2.2
100.0M
2.7B
5.4B
8.0B
Active Parameters


Compute budget: 3e+20 | Total Parameters
MoE Sparsity (sparser=darker)
0% (N: 1.3B)
25% (N: 1.8B)
50% (N: 2.2B)
75% (N: 3.4B)
90% (N: 6.6B)
95% (N: 9.6B)
98% (N: 18.8B)
Loss (validation)
3.4
3.2
3.0
2.8
2.6
2.4
2.2
100.0M
2.7B
5.4B
8.0B
Total Parameters


Compute budget: 1e+21 | Active Parameters
MoE Sparsity (sparser=darker)
0% (N: 2.9B)
25% (N: 2.6B)
50% (N: 2.4B)
75% (N: 1.8B)
90% (N: 1.5B)
95% (N: 1.3B)
Loss (validation)
3.0
2.8
2.6
2.4
2.2
100.0M
16.7B
33.4B
50.0B
Active Parameters


Compute budget: 1e+21 | Total Parameters
MoE Sparsity (sparser=darker)
0% (N: 2.9B)
25% (N: 3.3B)
50% (N: 4.4B)
75% (N: 6.1B)
90% (N: 12.1B)
95% (N: 21.2B)
Loss (validation)
3.2
3.0
2.8
2.6
2.4
2.2
100.0M
16.7B
33.4B
50.0B
Total Parameters